

# $G \times S$ : Decoupling *How* to Grasp from *What* to Grasp for Language-Driven Grasp Detection

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<https://github.com/RyleHan/gxs-grasp>

## Abstract

*Language-driven grasp detection asks for a grasp rectangle that satisfies a natural-language instruction. Current methods inject the instruction into the whole grasp network, so the text also has to shape the grasp geometry, which it does not determine. We factorise the grasp quality map into a language-agnostic graspability map  $G$  and an instruction-conditioned selection map  $S$ .  $G$  is a GR-ConvNet trained on the union of all grasps annotated in an image;  $S$  compares frozen dense CLIP features with the instruction and only has to choose among graspable regions. Both are supervised with labels that Grasp-Anything++ already provides: every image carries several instructions, and their grasps act as each other’s negatives, so no masks are needed. On the official test split  $G \times S$  raises the success rate from  $[x]$  to  $[y]$ , and the rate at which both grasps of an image pair are correct from  $[a]$  to  $[b]$ .*

## 1. Introduction

A robot asked to “grip the keychain by its ring” has to answer two questions: which part of the image the sentence refers to, and how a parallel gripper should close on it. Grasp-Anything++ [8] made this task learnable at scale, pairing about a million synthetic scenes with grasp instructions. Methods trained on it [7, 8] condition the entire detector on the text.

Two properties of the data suggest a simpler structure. First, the opening width and angle of a grasp at a given pixel follow from the object’s shape; the sentence only decides *where* to grasp. Second, instructions come in groups: one image carries a grasp instruction for each annotated object, and these point to different regions. The union of their grasps says where the image can be grasped at all; their differences say what each sentence adds.

We build on both observations.  $G \times S$  predicts  $Q = G \cdot S$ , where  $G$  ignores the language and  $S$  selects among the graspable regions. Our contributions are: (i) a factorised detector whose geometry heads never see the text; (ii) a sibling-contrast loss that supervises selection with the other

instructions of the same image, without any masks; and (iii) a paired metric that a model ignoring the instruction cannot pass, together with an analysis of the data showing that most part-level labels do not change with the part named in the instruction.

## 2. Related Work

**Language-driven grasping.** CLIPort [5] fuses CLIP [4] features into dense pick-and-place affordances. CROG [6] extends CLIP-based referring segmentation [10] with grasp maps. On Grasp-Anything++ [8, 9], LGD casts grasp detection as conditional diffusion, and mask-guided methods [1, 7] use object masks to focus the grasp features. Unlike these, we multiply at the *output* rather than the feature level, keep geometry language-free, and need no mask annotations.

**Dense CLIP features.** CLIP patch tokens are poorly localised; MaskCLIP [11] uses the value embedding of the last attention layer and ClearCLIP [3] drops its residual and feed-forward paths. We adopt this combination without training.

## 3. Method

Given an RGB image  $I$  and an instruction  $\ell$ , we predict  $g = (x, y, w, h, \theta)$ .

**Graspability  $G$ .** A GR-ConvNet [2] maps  $I$  to five  $224 \times 224$  maps: quality logits  $q$ ,  $\cos 2\theta$ ,  $\sin 2\theta$  (a parallel gripper is symmetric under  $180^\circ$ ), opening  $w$  and jaw size  $h$ . None of them receive  $\ell$ . For an image with instructions  $\ell_1, \dots, \ell_K$  and grasp sets  $\mathcal{G}_1, \dots, \mathcal{G}_K$ , targets are rendered from  $\bigcup_k \mathcal{G}_k$ :

$$\mathcal{L}_G = \text{BCE}(q, \hat{q}) + \sum_{m \in \{\cos, \sin, w, h\}} \|(m - \hat{m}) \odot \hat{q}\|_{\text{sL1}}, \quad (1)$$

where  $\hat{q}$  marks the centre third of each rectangle. Supervising geometry only where a grasp exists matters: with GR-ConvNet’s dense regression the predicted opening collapsed to  $\sim 15$  px ( $\sim 80$  px ground truth) even when overfitting 70 scenes.

**Selection  $S$ .** Frozen CLIP ViT-B/16 at  $448 \times 448$  gives  $28 \times 28$  patch features  $v_p$  (value-only last block);  $t$  is the CLIP text embedding. With zero-initialised residual adapters  $\phi, \psi$ ,

$$S(p | \ell) = \sigma(\tau \langle \phi(v_p), \psi(t) \rangle + b). \quad (2)$$

For instruction  $\ell_k$ , patches covered by  $\mathcal{G}_k$  are positives, patches covered by a sibling  $\mathcal{G}_{j \neq k}$  are negatives, and the rest are ignored, since background is already suppressed by  $G$ . We use a class-balanced BCE over these patches.

**Inference.**  $Q = \sigma(q) \cdot S \uparrow$  ( $S$  bilinearly upsampled); the grasp centre is  $\arg \max Q$  and  $\theta, w, h$  are read from  $G$ 's maps at that pixel. The backbone runs once per image; a new instruction costs one dot product.

## 4. Experiments

**Data.** We keep the official LGD test splits (seen / unseen categories). The official training split contains only **[n]** multi-object scenes, so we add **[n]** multi-object scenes from the full release, requiring that (R1) no scene appears in a test split, (R2) every object name belongs to the official training vocabulary, and (R3) no two objects share a name. In the resulting training set, **[r]**% of part pairs of the same object have identical grasp labels, so language mostly has to pick the object.

**Metrics.** *Success:*  $\text{IoU} > 0.25$  and  $|\Delta\theta| < 30^\circ$  with a ground-truth grasp of the instruction [8]. *Paired:* for each pair of objects in a multi-object test image, both grasps must succeed; ignoring the instruction gives almost zero. *Select:* the predicted centre lies on the target object.

**Results.** **[table + discussion]**

## 5. Conclusion

Separating how to grasp from what to grasp lets each part learn from the supervision it actually needs. The selection map inherits CLIP's open vocabulary, while geometry is learned once per image. The main limitation is CLIP's coarse  $28 \times 28$  grid, which limits part-level selection; the labels of Grasp-Anything++ rarely distinguish parts anyway.

## References

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